

Experiment No. 20

Write a Prolog Program to implement Towers of Hanoi

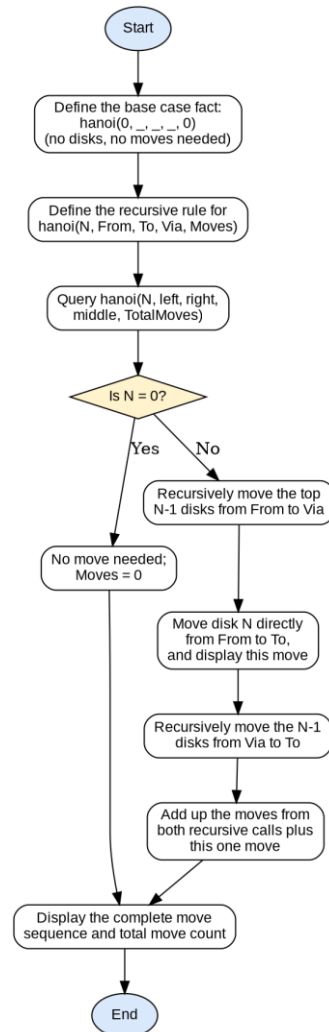
Aim

To write a Prolog program that solves the Towers of Hanoi puzzle for n disks, printing each move required to transfer all disks from the source peg to the destination peg using an auxiliary peg.

Algorithm

1. Start.
2. Define the base case fact: `hanoi(0, _, _, 0)`, meaning moving zero disks needs zero moves.
3. Define the recursive rule for `hanoi(N, From, To, Via, Moves)`: to move N disks from From to To using Via as the auxiliary peg.
4. Query `hanoi(N, left, right, middle, TotalMoves)` for the given number of disks N.
5. If $N = 0$, no move is needed and $Moves = 0$.
6. Otherwise, recursively move the top N-1 disks from From to Via (using To as the auxiliary).
7. Move disk N directly from From to To, and display this move.
8. Recursively move the N-1 disks from Via to To (using From as the auxiliary).
9. Add up the moves from both recursive calls plus this one move to get the total.
10. Display the complete sequence of moves and the total number of moves used.
11. Stop.

Flowchart



Program

```

hanoi(0, _, _, _, 0) :- !.
hanoi(N, From, To, Via, Moves) :-
    N > 0,
    N1 is N - 1,
    hanoi(N1, From, Via, To, M1),
    format("Move disk ~w from ~w to ~w~n", [N, From, To]),
    hanoi(N1, Via, To, From, M2),
    Moves is M1 + M2 + 1.

main :-
    N = 3,
    format("Towers of Hanoi solution for ~w disks~n", [N]),
    format("-----~n"),

```

```

hanoi(N, left, right, middle, TotalMoves),
format("-----~n"),
format("Total moves used: ~w~n", [TotalMoves]).

```

Result

The program was executed successfully and the output was verified as correct.

```
?- main.
```

Towers of Hanoi solution for 3 disks

```

-----
Move disk 1 from left to right
Move disk 2 from left to middle
Move disk 1 from right to middle
Move disk 3 from left to right
Move disk 1 from middle to left
Move disk 2 from middle to right
Move disk 1 from left to right
-----
Total moves used: 7
true.

```

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

```

hanoi.pl
1 hanoi(0, _, _, _, 0) :- !.
2 hanoi(N, From, To, Via, Moves) :-
3   N > 0,
4   N1 is N - 1,
5   hanoi(N1, From, Via, To, M1),
6   format("Move disk ~w from ~w to ~w~n", [N, From, To]),
7   hanoi(N1, Via, To, From, M2),
8   Moves is M1 + M2 + 1.
9
10 main :-
11   N = 3,
12   format("Towers of Hanoi solution for ~w disks~n", [N]),
13   format("-----~n"),
14   hanoi(N, left, right, middle, TotalMoves),
15   format("-----~n"),
16   format("Total moves used: ~w~n", [TotalMoves]).

```

```

PS C:\Users\Athavan\AI-Lab> swipl hanoi.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.
Towers of Hanoi solution for 3 disks
-----
Move disk 1 from left to right
Move disk 2 from left to middle
Move disk 1 from right to middle
Move disk 3 from left to right
Move disk 1 from middle to left
Move disk 2 from middle to right
Move disk 1 from left to right
-----
Total moves used: 7
true.
?-

```